Martin Daigne

Robotics Master's Student — Data Science Minor

martin.daigne@epfl.ch | 06 80 60 18 35 | Lausanne, Switzerland | LinkedIn | GitHub | Website

Education

MSc **École Polytechnique Fédérale de Lausanne (EPFL)**, Robotics - Data Science minor — Switzerland, 2024 – 2027
- Core courses: Machine Learning, Deep Learning, Reinforcement Learning, Convex Optimization, Model Predictive Control, Learning and Adaptive Control for Robots, Mobile Robotics, Robotics for Manipulation, Applied Data Analysis, Modern Natural Language Processing…

Delft University of Technology (TU Delft), Exchange - Mechanical/Aerospace Engineering — Netherlands, 2023 – 2024

BSc **École Polytechnique Fédérale de Lausanne (EPFL)**, Mechanical Engineering — Switzerland, 2021 – 2024
- Grade: 5.57/6
- Core courses: Analysis, General Physics (Mechanics, Thermodynamics, Electromagnetics), Numerical Analysis, Dynamical Systems, Continuum Mechanics, Structural Mechanics, Nonlinear Mechanics, Fluid Mechanics, Experimental Fluids, Heat and Mass Transfer, Control Systems, MEMS, Intelligent Vehicles…

Experience

EPFL, Laboratory of Intelligent Systems (LIS), Semester Project — Switzerland, Feb 2026 – June 2026, 5 months
- Design and implementation of diffusion policies (DDPM, DDIM, Score Matching, Flow Matching) to generate control trajectories for a morphing drone previously controlled with an MPPI.

EPFL, Computational Robot Design and Fabrication Lab (CREATE), Semester Project — Switzerland, Sept 2025 – Jan 2026, 5 months
- Data-Enabled Model Predictive Control for Compliant Manipulators: I built a simulation in Python and an adaptive MPC controller for a soft, tendon-driven arm using CasADi and acados for real-time deployment.

EPFL, Unsteady Flow Diagnostics Laboratory (UNFoLD), Summer in the Lab Internship — Switzerland, June 2024 – Aug 2024, 3 months
- Design, building, and analysis of a random gust generator to improve realism in wind turbine experiments.

Geneva University (UNIGE), Athena Program, Student Researcher — Switzerland, Sept 2020 – June 2021, 10 months
- 1st semester: Set Theory and Logic in Mathematics alongside high school.
- 2nd semester: Research on Kuratowski's Theorem in Set Theory.

Aether Swiss Kite, Control Team Leader — Switzerland, 2025 – 2026, 1 year
- Lead the data acquisition, system identification, and control of the kite to achieve optimized automated control.

Aether Swiss Kite, Electrical and Control Engineer — Switzerland, 2024 – 2025, 1 year
- 1st semester: Design, building, and integration of a position sensor of the kite.
- 2nd semester: Data acquisition and filter implementation (EKF, UKF) to localize the kite.

EPFL Rocket Team, Recovery System Engineer — Switzerland, 2022 – 2023, 1 year
- Design and testing of the parachute of the Nordend rocket of the EPFL Rocket Team for the European EUROC competition.

EPFL, Teaching Assistant — Switzerland

- Teaching assistant in Introduction to Computer Science, Analysis II, Fluid Mechanics, and Model Predictive Control. — 2022–2023 / 2024–2025

Awards

Deep Learning Object Detection Competition — 2025

Winner of the Deep Learning Object Detection competition 2025 in the Image Analysis and Pattern Recognition course at EPFL.

EPFL Space Race — 2022

Winner EPFL Space Race 2022.

Certificates

Cambridge English Advanced Certificate — 2020

English, C1 level.

Goeth Institut — 2018

German B1

Skills

Coding Languages: Strong experience in Python (PyTorch, Pandas, scikit-learn, …). Worked on Matlab and knowledge in C/C++ (mostly for Arduino).

Languages

French: Mother tongue

English: C1

Dutch: Fluent

German: Scholar